TATHAGAT

AI Engineer • LLM Systems • Computer-Vision Inference • Backend Engineering

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Open to: Remote · Hybrid · On-site · Relocation

PROFESSIONAL PROFILE

AI/ML Engineer with 2.5+ years of hands-on experience building production-oriented Generative AI, RAG, agentic AI, computer vision, and healthcare SaaS systems. Strong in Python, FastAPI, Hugging Face, Qwen/vLLM, Docker, PostgreSQL, Redis, vector search, and AWS. Experienced across the model lifecycle: experimentation, domain-specific fine-tuning, inference serving, API integration, evaluation-oriented feedback loops, structured logging, deployment planning, and production hardening. Brings a pragmatic health-tech mindset grounded in PHI-aware access control, auditability, reliable delivery, and safe system behavior. Portfolio with case studies and interactive experiments: tathagat.xyz.

PROFESSIONAL EXPERIENCE

Technical Director, AI / Lead AI Engineer • Vigyanam Labs **Dec 2025 – Present**

Mohali, India

- Own end-to-end AI architecture and delivery for startup product initiatives, translating ambiguous business needs into PoCs, ML/LLM services, technical roadmaps, API contracts, implementation tasks, and release plans.
- Design production-oriented systems across Generative AI, RAG, computer vision, OCR, healthcare workflows, model serving, secure APIs, and cloud deployment; collaborate with backend, frontend, DevOps, CloudOps, and client stakeholders.
- Lead architecture reviews, debugging, test planning, documentation, and production hardening—balancing research exploration with dependable sprint delivery.

AI/ML Engineer — Netsmartz **Jul 2024 – Dec 2025**

Mohali, India

- Built AI/ML applications spanning modular RAG, LLM fine-tuning, forecasting, invoice extraction, and computer vision, with production-style data pipelines and API integrations.
- Developed FastAPI AI backends using PostgreSQL, Redis, Milvus/FAISS, async routes, authentication, feedback capture, Dockerized services, structured logging, and external LLM/API integrations.
- Fine-tuned and served domain-specific LLMs using custom datasets and Hugging Face tooling; designed high-concurrency inference approaches around local model servers and vLLM-compatible endpoints.
- Led client-facing architecture discussions, PoC planning, debugging sessions, implementation handoffs, and cross-team delivery for AI and backend initiatives.

Data Analyst Intern — Salt Holdings **Jan 2024 – Jul 2024**

Mohali, India

- Supported analytics and ML workflows with Python and SQL, including data cleaning, exploratory analysis, feature engineering, model-ready dataset preparation, and business reporting.

KEY PROJECTS

FacialDx — AI Clinical Inference System 2025 – Present

Python · FastAPI · YOLO · EfficientNet · Claude Vision · asyncio · AWS EC2/ALB/S3/Aurora

- Stateless, horizontally scalable AI inference for PTSD, TBI, and depression detection — 50+ endpoints built for reliability, concurrency, and screening throughput.
- Concurrent inference via asyncio.gather (YOLO + EfficientNet + Claude Vision); Uvicorn multi-worker GPU inference on AWS ALB with Prometheus monitoring.
- Production-grade auth (LRU-cached Secrets Manager), SlowAPI rate limiting, and async PostgreSQL (SQLAlchemy + asyncpg).
- Built healthcare workflows for multi-image facial-feature analysis and structured report generation; supported model debugging, inference planning, and separation of GPU-heavy execution from core API handling.
- Designed secure B2B tenancy, organization/clinic/staff/patient flows, RBAC, audit logs, and PHI-aware image storage to support traceable healthcare operations.
- Led the production hardening and model-parity work for a GPU-backed Triton inference service, validating ONNX/TensorRT-style model outputs against the legacy pipeline, debugging image-preprocessing and detection differences, and improving reliable deployment of high-throughput AI analysis workloads.

Pronto CodeLLM — Agentic Chatbot on Proprietary Coding Language 2024 – 2025

Qwen · Llama · Phi · LLM Fine-tuning · LoRA/Adapter · Agentic RAG · Local GPU (2× A100)

- Built an agentic RAG system that explains areas of Pronto's product codebase written in their proprietary language — enabling developers to query and understand code structure in natural language.
- Fine-tuned Qwen, Llama, and Phi models on the proprietary codebase to deeply understand language syntax, patterns, and code structure for accurate explanation generation.
- Trained a separate generation adapter (LoRA-based) on top of the fine-tuned model to enable text-to-code synthesis — converting natural-language instructions into structured proprietary code.
- Deployed on-premise on dual A100 GPUs for data privacy; no proprietary code left the client's infrastructure.

KEY PROJECTS (CONTINUED)

Mahindra Swaraj Tractors R&D — Should-Be-Costing and Gearbox Fault Detection

Computer Vision · OCR · YOLO · Gemini Vision · RAG · Signal Processing · ResNet-50 · EfficientNet · ViT

Gearbox Fault Detection

- Raw .wav acoustic data (and potentially signal/graph representations), with noise isolation and suppression of human/environmental noise.
- Signal processing: band-pass/Butterworth filtering, windowing, envelope spectrum, cepstrum, and spectral kurtosis.
- Gear-domain analysis: gear-mesh frequency (GMF) and harmonics, shaft-frequency sidebands, plus tachometer-based order tracking / synchronous averaging where RPM varied.
- Time-frequency features using Continuous Wavelet Transform (Morlet) scalograms.
- Deep-learning comparison using ResNet-50, EfficientNet, and Vision Transformer (ViT) on the scalograms; numerical signal features could be combined with the vision model.
- Explainability through Grad-CAM and SHAP.
- Fault targets: healthy, humming, rumble, and chatter—with rumble associated with bearing impacts/resonance and chatter with broadband impulsive tooth impacts.

Gear Should-Be-Costing

- Input: JPG/PNG exports of AutoCAD drawings for single gears.
- Drawing understanding: image preprocessing with OpenCV—denoising, binarisation, and deskewing—followed by geometry extraction using Canny/Hough lines or LineTR.
- Dimension and text extraction: OCR for drawing annotations, material, tolerances, and dimensions.
- Symbol/feature detection: YOLOv5 for manufacturing symbols and drawing elements.
- Structured feature output: gear teeth, module, outer/pitch/root diameters, face width, bore, material, and tolerance information.
- Knowledge retrieval: RAG over Milvus/FAISS to retrieve master data, historical should-cost examples, process information, and machine/rate references.
- Cost calculation: a deterministic engine using client-provided formulas and rates.

- Raw-material cost = estimated weight × material rate; operation cost = machining time × machine/labour rate; plus tooling and overhead.
- Output: an Excel should-cost sheet with separate raw-material cost and row-wise manufacturing-operation costs, kept under human validation.

TECHNICAL SKILLS

Agentic AI & LLMs: LangChain, LangGraph, AutoGen, CrewAI, OpenAI GPT-4o, Anthropic Claude 3.5, Llama 3, Qwen, Phi, Mistral, LlamaIndex

LLM Fine-tuning: LoRA, QLoRA, PEFT, Adapter training, Qwen/Llama/Phi fine-tuning, domain-specific NLP, Hugging Face, local GPU (A100)

RAG & Vector Data: Pinecone, FAISS, Weaviate, Chroma, Azure AI Search — RAG architecture, semantic search, embedding & chunking optimization

Backend & APIs: Python, FastAPI, asyncio, REST APIs, PostgreSQL, Microservices, Docker, CI/CD

LLMOps & Evaluation: LangSmith, Weights & Biases, MLflow, hallucination testing, LLM-as-a-Judge, prompt versioning, drift detection

Cloud & Infrastructure: AWS (EC2, ALB, S3, Aurora, Secrets Manager), Azure (AI Foundry, Document Intelligence, AI Search, SQL), IBM Cloud, Uvicorn, GPU inference

Automation & Integration: n8n, Stripe, Google Sheets API, Notion API, HubSpot

Machine Learning: LightGBM, XGBoost, Scikit-learn, ARIMA, NLP, Transformers, Hugging Face, Time Series Forecasting

Generative / Multimodal: Gemini, YOLO, EfficientNet, Claude Vision, fine-tuning, multimodal orchestration

EDUCATION

Master's in Data Science and Artificial Intelligence
Himachal Pradesh University, India · Research-focused AI/ML coursework and applied project work

Bachelor of Technology (B.Tech) — Information Technology
University Institute of Information Technology, Himachal Pradesh University, India

LANGUAGES

English — Fluent (C1) · Hindi — Native